

WASP-104b

R-2634

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_150406 · created 2026-08-29T00:43:29+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457118.80564 +/- 0.00054 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.044
Transit depth (Rp/Rs)^2	2.26 +/- 1.32 [%]
Orbital Inclination (inc)	83.37 +/- 2.83
Airmass coefficient 1 (a1)	0.37 +/- 0.21
Airmass coefficient 2 (a2)	0.67 +/- 0.3
Scatter in the residuals of the lightcurve fit is	52.83 %
Best Comparison Star	None
Optimal Aperture	1.46
Optimal Annulus	3.25
Transit Duration (day)	0.067 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.15 ± 0.044 vs 0.1214 (deviation 0.62σ)
Duration fit vs published	0.067 d vs 0.0742 d (deviation 0.24σ)
Mid-time O–C	-1.3 min
Depth SNR	3.2
Residual scatter	52.83 %

Light curve

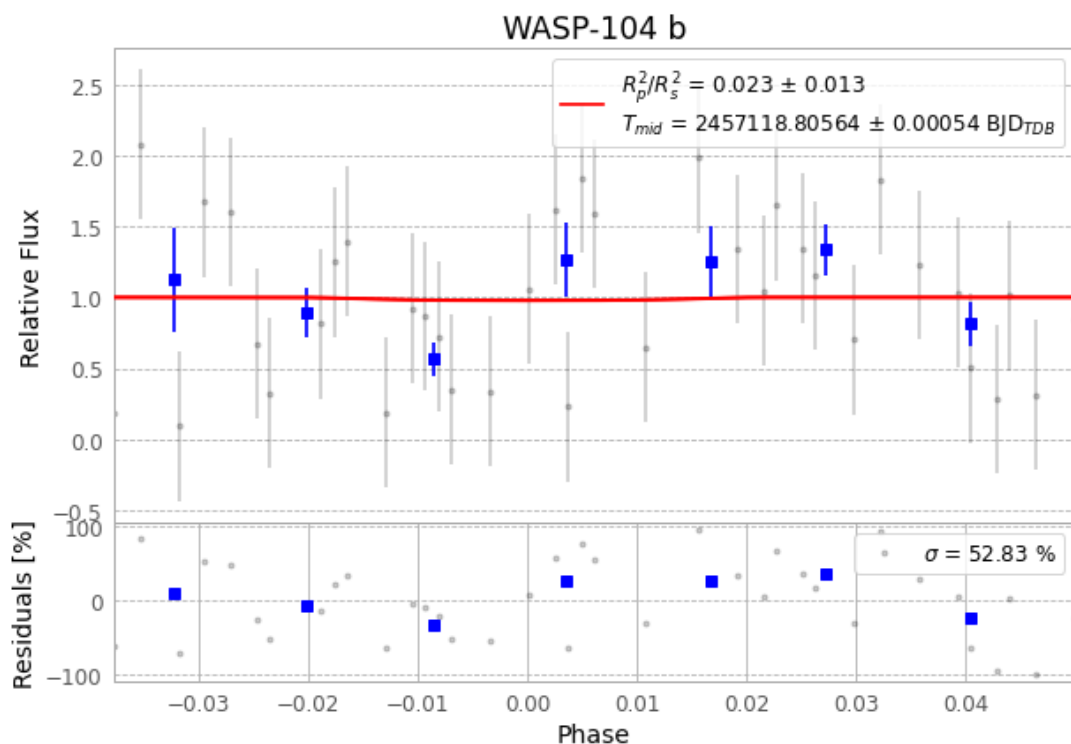


Figure 1 — FinalLightCurve_WASP-104 b_2015-04-05.png (SHA-256 9e387e8c7471536c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [155, 301], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fbb5e6fd2c975219...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

76 FITS frames (50 MB), WASP-104150406053316.FITS → WASP-104150406091812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-150406.log (54942f4a656f...), FinalLightCurve_WASP-104 b_2015-04-05.png (9e387e8c7471...), FinalLightCurve_WASP-104 b_2015-04-05.csv (bafbc9eee49f...)

Compute resources

Wall time 117.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 00:43Z.